

Q1 - 25 July - Shift 1

A unit scale is to be prepared whose length does not change with temperature and remains 20 cm, using a bimetallic strip made of brass and iron each of different length. The length of both components would change in such a way that difference between their lengths remains constant. If length of brass is 40 cm and length of iron will be _____ cm.

($\alpha_{\text{iron}} = 1.2 \times 10^{-5} \text{ K}^{-1}$ and $\alpha_{\text{brass}} = 1.8 \times 10^{-5} \text{ K}^{-1}$).

Space for your notes:

Q2 - 25 July - Shift 2

A block of ice of mass 120 g at temperature 0°C is put in 300 gm of water at 25°C . The xg of ice melts as the temperature of the water reaches 0°C . The value of x is

[Use: Specific heat capacity of water = $4200 \text{ J kg}^{-1} \text{ K}^{-1}$, Latent heat of ice = $3.5 \times 10^5 \text{ J kg}^{-1}$]

Space for your notes:

Q3 - 26 July - Shift 2

An ice cube of dimensions $60 \text{ cm} \times 50 \text{ cm} \times 20 \text{ cm}$ is placed in an insulation box of wall thickness 1 cm. The box keeping the ice cube at 0°C of temperature is brought to a room of temperature 40°C . The rate of melting of ice is approximately: (Latent heat of fusion of ice is $3.4 \times 10^5 \text{ J kg}^{-1}$ and thermal conducting of insulation wall is $0.05 \text{ W m}^{-1} \text{ }^\circ\text{C}^{-1}$)

- (A) $61 \times 10^{-1} \text{ kg s}^{-1}$ (B) $61 \times 10^{-5} \text{ kg s}^{-1}$
(C) 208 kg s^{-1} (D) $30 \times 10^{-5} \text{ kg s}^{-1}$

Space for your notes:

Q4 - 27 July - Shift 1

Questions

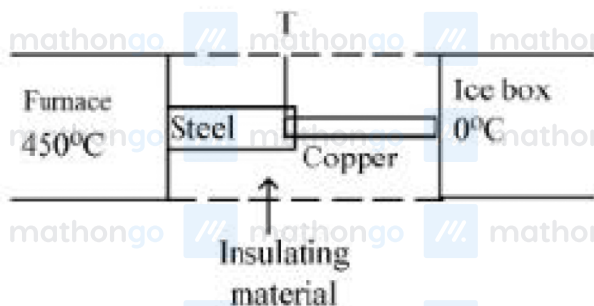
MathonGo

If K_1 and K_2 are the thermal conductivities L_1 and L_2 are the lengths and A_1 and A_2 are the cross sectional areas of steel and copper rods

respectively such that $\frac{K_2}{K_1} = 9, \frac{A_1}{A_2} = 2, \frac{L_1}{L_2} = 2$.

Then, for the arrangement as shown in the figure.

The value of temperature T of the steel – copper junction in the steady state will be :



(A) 18°C

(B) 14°C

(C) 45°C

(D) 150°C

Space for your notes:

Q5 - 27 July - Shift 1

Questions

MathonGo

Read the following statements :

Space for your notes:

A. When small temperature difference between a liquid and its surrounding is doubled the rate of loss of heat of the liquid becomes twice.

B. Two bodies P and Q having equal surface areas are maintained at temperature 10°C and 20°C . The thermal radiation emitted in a given time by P and Q are in the ratio 1 : 1.15

C. A carnot Engine working between 100 K and 400 K has an efficiency of 75%

D. When small temperature difference between a liquid and its surrounding is quadrupled, the rate of loss of heat of the liquid becomes twice.

Choose the correct answer from the options given below :

(A) A, B, C only

(B) A, B only

(C) A, C only

(D) B, C, D only

Q6 - 29 July - Shift 1

Two metallic wires of identical dimensions are connected in series. If σ_1 and σ_2 are the conductivities of the these wires respectively, the effective conductivity of the combination is :

Space for your notes:

(A) $\frac{\sigma_1 \sigma_2}{\sigma_1 + \sigma_2}$

(B) $\frac{2\sigma_1 \sigma_2}{\sigma_1 + \sigma_2}$

(C) $\frac{\sigma_1 + \sigma_2}{2\sigma_1 \sigma_2}$

(D) $\frac{\sigma_1 + \sigma_2}{\sigma_1 \sigma_2}$

Q7 - 29 July - Shift 2

#MathBoleTohMathonGo

Nearly 10% of the power of a 110 W light bulb is converted to visible radiation. The change in average intensities of visible radiation, at a distance of 1 m from the bulb to a distance of 5 m is $a \times 10^{-2} \text{ W/m}^2$. The value of 'a' will be

Space for your notes:

Answer Key**Q1 (60)****Q2 (90)****Q3 (B)****Q4 (C)****Q5 (A)****Q6 (B)****Q7 (84)**

#MathBoleTohMathonGo

Q1 (60)

$$\ell_B(1 + \alpha_B \Delta T) - \ell_i(1 + \alpha_i \Delta T) = \ell_B - \ell_i$$

$$\alpha_B \ell_B = \ell_i \alpha_i$$

$$1.8 \times 10^{-5} \times 40 = \ell_i \times 1.2 \times 10^{-5}$$

$$\ell_i = \frac{1.8 \times 10^{-5} \times 40}{1.2 \times 10^{-5}} = \frac{3 \times 40}{2} = 60$$

$$\ell_i = 60 \text{ cm}$$

Q2 (90)

Energy released by water

$$= 0.3 \times 25 \times 4200 = 31500 \text{ J}$$

let m kg ice melts

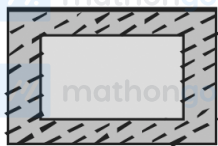
$$m \times 3.5 \times 10^5 = 31500$$

$$m = \frac{31500 \times 10^{-5}}{3.5} = 9000 \times 10^{-5}$$

$$m = 0.09 \text{ kg} = 90 \text{ gm}$$

$$x = 90$$

Q3 (B)



$$\frac{dQ}{dt} = \frac{KA\Delta T}{\ell}$$

$$A = 2(0.6 \times 0.5 + 0.5 \times 0.2 + 0.2 \times 0.6)$$

$$= 2(0.3 + 0.1 + 0.12)$$

$$= 2(0.4 + 0.12)$$

$$= 2(0.52)$$

$$= 1.04 \text{ m}^2$$

$$R_{th} = \frac{\ell}{KA} \Rightarrow \frac{1 \times 10^{-2}}{0.05 \times 1.04} \Rightarrow \frac{10^{-2}}{0.052}$$

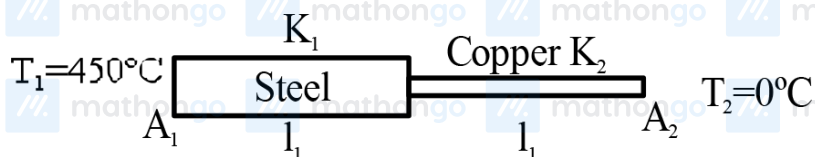
$$\frac{dQ}{dt} = \frac{\Delta T}{R_{th}} \Rightarrow \frac{40 \times 0.052}{10^{-2}} \Rightarrow 2.08 \times 10^2 \text{ J/s}$$

$$2.08 \times 10^2 = m \times 3.4 \times 10^5$$

$$m = \frac{2.08}{3.4 \times 10^3} \Rightarrow 0.61 \times 10^{-3} \text{ kg/s}$$

$$= 61 \times 10^{-5} \text{ Kg/s}$$

Q4 (C)



$$\frac{d\theta}{dt} = \frac{K_1 A_1}{l_1} (T_1 - T) = \frac{K_2 A_2}{l_2} (T - T_2)$$

$$\Rightarrow \frac{450 - T}{T - 0} = \frac{K_2 A_2 l_1}{K_1 A_1 l_2} = 9 \times \frac{1}{2} \times 2$$

$$\Rightarrow 450 - T = 9T \Rightarrow T = 45^\circ\text{C}$$

Q5 (A)

Heat Transfer

A. by Newton's law of cooling $\frac{d\theta}{dt} \propto \Delta T$

$$\text{B. } H = \frac{d\theta}{dt} = \sigma e A T^4 \Rightarrow \frac{H_P}{H_Q} = \left(\frac{T_P}{T_Q} \right)^4 = \left(\frac{283}{293} \right)^4$$

$$H_P : H_Q = 1 : (1.03)^4 = 1 : (1.03)^4 = 1 : 1.15$$

 $\Rightarrow$ B is correct

$$\text{C. } \eta = 1 - \frac{100}{400} = \frac{3}{4} = 75\%$$

D. is wrong as $\frac{d\theta}{dt} \propto \Delta T$ **Q6 (B)**

$$A \left(\begin{array}{c} \sigma_1 \\ \ell \end{array} \right) \left(\begin{array}{c} \sigma_2 \\ \ell \end{array} \right) \equiv \left(\begin{array}{c} \sigma_{eq} \\ 2\ell \end{array} \right) A$$

Let length of wire be ' ℓ '

Area of wire as ' A '

For equivalent wire length = 2ℓ & area will be A

Thermal resistance

$$R_{eq} = R_1 + R_2$$

$$\frac{2\ell}{\sigma_{eq} A} = \frac{\ell}{\sigma_1 A} + \frac{\ell}{\sigma_2 A}$$

$$\frac{2\ell}{\sigma_{eq}} = \frac{\ell}{\sigma_1} + \frac{\ell}{\sigma_2} \Rightarrow \sigma_{eq} = \frac{2\sigma_1\sigma_2}{\sigma_1 + \sigma_2}$$

Q7 (84)

$$P' = 10\% \text{ of } 110 \text{ W}$$

$$= \frac{10}{100} \times 110 \text{ W}$$

$$= 11 \text{ W}$$

$$I_1 - I_2 = \frac{P'}{4\pi r_1^2} - \frac{P'}{4\pi r_2^2}$$

$$= \frac{11}{4\pi} \left[\frac{1}{1} - \frac{1}{25} \right]$$

$$= \frac{11}{4\pi} \times \frac{24}{25}$$

$$= \frac{264}{\pi} \times 10^{-2} = 84 \times 10^{-2} \text{ W / m}^2$$